# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## CLASS 10 — SCIENCE TWO-CHAPTER TEST SET B

Chapters 1 & 2: **Chemical Reactions and Equations, Acids, Bases and Salts** | RBSE Board Pattern | Session 2026–27

**Time: 1 Hour 30 Minutes** Name: _____ Roll No.: _____ **Maximum Marks: 40**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

## SECTION — A (Objective Type Questions) 10 × 1 = 10

*Choose the correct option and write it in your answer book.*

**1.** $2\,Na + 2\,H_2O \longrightarrow 2\,NaOH + H_2$ is a: 1

**(a)** decomposition reaction **(b)** double displacement reaction
**(c)** displacement reaction **(d)** combination reaction

**2.** In $CuO + H_2 \longrightarrow Cu + H_2O$, Student A says CuO is reduced; Student B says $H_2$ is reduced. Then: 1

**(a)** Student A is correct **(b)** Student B is correct
**(c)** Both are correct **(d)** Both are wrong

**3.** Decomposition of vegetable matter into compost is an example of: 1

**(a)** a displacement reaction **(b)** a physical change
**(c)** an endothermic reaction **(d)** an exothermic reaction

**4.** Antioxidants are added to food containing fats and oils to prevent: 1

**(a)** rancidity **(b)** corrosion **(c)** precipitation **(d)** fermentation

**5.** The symbol (aq) in a chemical equation means: 1

**(a)** aqueous solution **(b)** acid **(c)** air **(d)** aquatic

**6.** Student A says vinegar contains acetic acid; Student B says it contains citric acid. Then: 1

**(a)** Student B is correct **(b)** Student A is correct
**(c)** Both are correct **(d)** Both are wrong

**7.** The pH of a solution with $OH^-$ concentration higher than $H^+$ is: 1

**(a)** 0 **(b)** less than 7 **(c)** more than 7 **(d)** exactly 7

**8.** Which of the following is a strong base? 1

**(a)** copper hydroxide **(b)** ammonium hydroxide
**(c)** magnesium hydroxide **(d)** sodium hydroxide

**9.** The gas evolved when an acid reacts with a metal hydrogencarbonate is: 1

**(a)** $H_2$ **(b)** $Cl_2$ **(c)** $CO_2$ **(d)** $O_2$

**10.** Common salt is obtained from sea water by: 1

**(a)** sublimation **(b)** distillation **(c)** evaporation **(d)** filtration

## SECTION — B (Very Short Answer Type) 4 × 2 = 8

*Answer in about 10–15 words each.*

**11.** Define a precipitation reaction with an example. 2

**12.** A student says oily food becomes rancid due to reduction. Justify or refute. 2

**13.** Why does a solution of NaCl not change the colour of litmus? 2

**14.** A student says that acids and bases neutralise each other only in the laboratory. Justify or refute. 2

## SECTION — C (Short Answer Type) — 4 × 3 = 12

*Answer in about 30–40 words each.*

**15.** State three effects of oxidation reactions in everyday life with an example each. 3

**16.** Explain why 3
(i) antacid tablets relieve acidity
(ii) toothpaste is basic
(iii) lime is added to acidic soil.

**17.** Write balanced equations: 3
(i) potassium reacts with water
(ii) zinc carbonate is heated
(iii) carbon dioxide is passed through lime water.

**OR**

Why is the reaction of an acid with a base called double displacement? Write the equation for NaOH and HCl and state the ions exchanged. 3

**18.** How is washing soda obtained from baking soda? Write the equations. 3

**OR**

What happens to the concentration of $H_3O^+$ ions when excess base is added to an acid solution? What happens to the pH? Why? 3

## SECTION — D (Long Answer Type) — 2 × 5 = 10

*Answer in about 100–150 words each.*

**19.** (i) Write balanced equations with state symbols for: 5
(a) barium chloride + sodium sulphate solutions
(b) sodium hydroxide + hydrochloric acid
(c) zinc + dilute sulphuric acid.
(ii) Identify the type of each.
(iii) Which one forms a precipitate?

**OR**

**Case study:** A student burns a magnesium ribbon. 5
(i) What is the colour of the flame?
(ii) Name the product and state its nature when dissolved in water.
(iii) Write the two equations involved.
(iv) Why is the ribbon cleaned first?

**20.** (i) Write the equation for the reaction of sodium hydroxide with hydrochloric acid. 5
(ii) Describe an activity to show neutralisation using phenolphthalein.
(iii) Why is heat released?

**OR**

(i) What is gypsum? 5
(ii) How is Plaster of Paris made from it?
(iii) Write the equation for the setting of Plaster of Paris.
(iv) Name the medical use.

## END OF QUESTION PAPER — Total: 40 Marks

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